

⊗ CMP

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① a.

We have a periodic potential:

$$V(x) = \sum_g V_g e^{iGx} \quad \text{where}$$

$$\Rightarrow \text{Free } e^- \text{ energy: } E_0(k) = \frac{\hbar^2 k^2}{2m} \quad G = \frac{2\pi n}{a}$$

Wavefunctⁿ. in plane waves $\Rightarrow$

$$\psi_k(x) = \sum_G C_{k-G} e^{i(k-G)x}$$

By Schrodinger eqⁿ

$$\left(\frac{\hbar^2 (k-G)^2}{2m} - E \right) C_{k-G} + \sum_{G'} V_{G'} C_{k-G-G'} = 0$$

$\Rightarrow$ For weak potentials, only nearly degenerate plane waves mix.

$\Rightarrow$ At the center of BZ ($k=0$):

$$\text{Energy} \Rightarrow E_0(k) = \frac{\hbar^2 k^2}{2m}$$

$\downarrow$

Even for k : $E_0(k) = E_0(-k)$

$\therefore$ The hamiltonian is invariant under $k \rightarrow -k$;

so, $E(k) = E(-k)$

$\Rightarrow$ diff. w.r.t k :

$$\frac{dE}{dk} = -\frac{dE}{dk}$$

$$\Rightarrow \frac{dE}{dk} = 0 \quad \text{at } k=0$$

Hence, slope is zero at zone center.

$\Rightarrow$ At BZ boundary $k = \frac{G}{2}$:

At this point:

$$E_0\left(\frac{G}{2}\right) = \frac{\hbar^2}{2m} \left(\frac{G}{2}\right)^2$$

Checking degeneracy;

$$E_0\left(\frac{G}{2}\right) = E_0\left(\frac{G}{2} - G\right) = E_0\left(-\frac{G}{2}\right)$$

$\therefore$ The two plane waves are degenerate:

$$e^{-i\frac{G}{2}x} \quad \text{and} \quad e^{-i\frac{G}{2}x}$$

By two-state approximation,
we keep only 2 coefficients:

$$C_K \text{ \& } C_{K-G}$$

→ Matrix eqⁿ:

$$\begin{pmatrix} E_0(K) & V_G \\ V_G^* & E_0(K-G) \end{pmatrix} \begin{pmatrix} C_K \\ C_{K-G} \end{pmatrix} = E \begin{pmatrix} C_K \\ C_{K-G} \end{pmatrix}$$

§. The determinant condⁿ

$$\hookrightarrow (E_0(K) - E)(E_0(K-G) - E) - |V_G|^2 = 0$$

Now, we'll expand near $K = \frac{G}{2}$:

$$\text{Let } K = \frac{G}{2} + q$$

$$\Rightarrow E_0(K) = \frac{\hbar^2}{2m} \left(\frac{G}{2} + q \right)^2$$

$$E_0(K-G) = \frac{\hbar^2}{2m} \left(-\frac{G}{2} + q \right)^2$$

on expanding:

$$E_0(K) = E_0\left(\frac{G}{2}\right) + \frac{\hbar^2 G}{2m} q$$

$$E_0(K-G) = E_0\left(\frac{G}{2}\right) - \frac{\hbar^2 G}{2m} q$$

Now, putting in secular eqⁿ:

$$(E_0 - E + \alpha q)(E_0 - E - \alpha q) - |V_G|^2 = 0$$

$$\text{where, } \alpha = \frac{\hbar^2 G}{2m}$$

Thus; $(E_0 - E)^2 - \alpha^2 q^2 - |V_G|^2 = 0$

on solving we get;

$$E = E_0 \pm \sqrt{\alpha^2 q^2 + |V_G|^2}$$

Thus; $\frac{dE}{dk} = \frac{dE}{dq} = \pm \frac{\alpha^2 q}{\sqrt{\alpha^2 q^2 + |V_G|^2}}$

At Boundary; $q=0$

So, $\frac{dE}{dk} = 0 \Rightarrow$ Hence, slope is zero at BZ edge.

(b) To show band gap depends on Fourier harmonic strength.

At exact boundary, $q=0$

$$E = E_0 \pm |V_G|$$

upper band: $E_+ = E_0 + |V_G|$

Lower " : $E_- = E_0 - |V_G|$

Band Gap $\Rightarrow \Delta E = E_+ - E_-$

$$\Delta E = 2|V_G|$$

$\therefore V_G$ is the Fourier Coefficient of the periodic potential,

$$\Delta E = 2|V_G|$$

Thus band Gap magnitude is directly proportional to the strength of that harmonic.

i.e Higher Zone boundaries correspond to higher G ,
So Gap depends on corresponding V_G .
